

Lightweight AI-Based Intrusion Detection Models for IoT Devices: A Comparative Review

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Abstract— The rapid expansion of the Internet of Things (IoT) has introduced a vast, vulnerable attack surface composed of devices with strict constraints on computation, memory, and energy. Traditional host or network centric intrusion detection systems (IDS) often exceed these limitations, prompting over a decade of research into lightweight, AI driven alternatives. This review synthesizes over eighty peer reviewed studies from 2013 to 2024 that apply machine learning or deep learning techniques to IoT security. The works are grouped into three main design families: (i) anomaly detection models using shallow learners or autoencoders, (ii) tree based and other classical classifiers optimized via feature selection or heuristic tuning, and (iii) distributed and federated approaches that shift heavy computation to edge or fog layers. Models are evaluated on detection accuracy, false-alarm rate, CPU overhead, memory footprint, and energy use. Results show that compressed deep learning and hybrid ensembles can now exceed 97% detection rates while using under 15 MB of memory and <8% CPU on typical microcontroller platforms. However, challenges remain such as scarce labeled IoT attack data, concept drift, and lack of standardized evaluation frameworks across diverse hardware. The review concludes by highlighting future directions, including adaptive continual learning, privacy-preserving model updates, and lightweight IDS benchmarks, to help guide the development of deployable, resource efficient intrusion detection for emerging IoT systems.

Keywords— *IoT Security, Intrusion Detection, Lightweight Machine Learning, Resource-Constrained-Devices, Edge Computing, Anomaly Detection, Artificial Intelligence*

I. INTRODUCTION

The proliferation of the Internet of Things (IoT) has led to an unprecedented number of smart devices deployed in homes, industries, healthcare, transportation, and other domains. By 2025, the number of connected IoT devices is projected to exceed 75 billion globally [1]. These devices range from sensors and cameras to actuators and appliances, all generating and exchanging vast amounts of data. However, many IoT devices operate with minimal computational resources and limited memory/power and often lack robust built-in security [2][3]. This makes IoT ecosystems an attractive target for cyber attackers. High-profile incidents such as the Mirai botnet attack, which co-opted thousands of IoT devices to launch massive, distributed denial of service (DDoS) attacks, have underscored the severity of IoT security threats [4]. In this context, Intrusion Detection Systems (IDS) have become essential for IoT

environments as a line of defense that can monitor network and device activities to detect malicious behavior in real time. Traditional IDS solutions, however, are often too heavy to run on constrained IoT hardware. This has motivated significant research into lightweight, AI based intrusion detection techniques tailored for IoT devices [5].

A. Challenges in Intrusion Detection for IoT

Developing intrusion detection models for IoT devices presents several key challenges:

- **Resource Constraints:** IoT devices have low-power processors, limited memory, and constrained energy [6]. IDSs for IoT must operate within these limits, as complex algorithms or frequent monitoring can drain batteries or overwhelm CPUs. Recent measurements show certain ML models can significantly raise energy use on IoT hardware [7][15]. Ensuring IDSs are efficient and lightweight enough to run locally or semi-locally is a primary challenge [8].
- **Heterogeneity of Devices and Networks:** The IoT landscape is highly diverse, spanning many device types, operating systems, and protocols (Wi-Fi, Zigbee, 6LoWPAN, MQTT, etc.). A one-size-fits-all IDS is hard to achieve across this heterogeneity [2][9]. Attack patterns vary, for example, between industrial sensors and smart homes. This diversity complicates IDS design and data collection, as models must adapt to different contexts and threat models [5][10].
- **Data Scarcity and Evolving Threats:** Effective AI based IDSs need quality data for training and updates. In IoT, comprehensive labeled attack datasets are scarce, especially for new or sophisticated threats [11][12]. IoT attacks evolve continuously with new malware, and zero-day exploits can quickly outdated models. Thus, unsupervised or semi-supervised methods are needed to detect novel attacks, but maintaining accuracy over time is challenging [6][13]. IDSs must handle concept drift and continuously learn from new data without high false alarm rates.
- **Performance Trade-offs:** There is a trade-off between IDS detection performance and computational overhead. Complex models like deep neural networks can detect subtle attacks with high accuracy but may be

too slow or resource-heavy for IoT. Conversely, very simple or compressed models run efficiently but may miss complex attacks, leading to lower detection or more false negatives. Balancing this trade-off is challenging. Many studies optimize feature selection and dimensionality reduction to reduce model size while keeping accuracy [10][11][14]. Still, achieving low false positives and high detection with minimal resources remains difficult [9][11][15].

B. Objectives and Scope

This paper presents a comparative review of lightweight, AI-based intrusion detection models for IoT environments, focusing on their effectiveness under resource constraints. It examines how machine learning techniques balance detection accuracy with efficiency, and which strategies such as feature selection, model compression, and knowledge distillation enable practical deployment on limited hardware.



Figure 1 Objectives and Scope Process

The review spans IDS research from the past decade targeting constrained IoT and cyber-physical systems, with emphasis on anomaly-based methods for detecting unknown threats. AI-enhanced signature-based systems are also included. Covered approaches range from classical models (e.g., decision trees, SVMs) to neural networks and autoencoders, provided they are tailored for IoT. Following prior work [8], an IDS is “lightweight” if it detects threats without overloading CPU, memory, or energy. While edge and fog offloading are noted, the primary focus is on model-level design.

The paper is structured as follows: Section 2 covers IoT IDS background and related work; Section 3 reviews key lightweight models; Section 4 compares their performance and efficiency; Section 5 outlines open challenges and future directions; and Section 6 concludes. This analysis clarifies the state of lightweight AI driven IoT IDS and supports future research in this space.

II. BACKGROUND AND RELATED CONCEPTS

The rapidly expanding IoT ecosystem connects billions of resource-limited devices with cloud-scale analytics, creating a security landscape distinct from traditional networks. This section frames lightweight AI-based intrusion detection by outlining common IoT architectures and the security goals at each layer [16][29]. It surveys intrusion detection paradigms signature, anomaly, specification, and hybrid and their supervised and unsupervised learning methods [16]. Finally, it reviews optimization techniques that enable “TinyML” models to meet IoT’s strict memory, power, real-time, and heterogeneity demands, contrasting these with general-purpose endpoints.

A. IoT Architecture and Security Requirements

IoT deployments are often modeled as three layers: perception, network (or edge), and application [29]. The perception layer includes sensors and actuators with limited RAM and power, vulnerable to node capture, spoofing, and physical attacks [16]. The network layer uses low-power protocols like 6LoWPAN and MQTT but faces threats like DoS, MitM, and route manipulation [30]. The application/cloud layer risks code injection, malware, and data exfiltration [30].

Expanded models add middleware or processing tiers to offload computation and enforce security before cloud access, bridging edge cloud gaps while maintaining visibility [29]. For example, SVELTE embeds minimal inspection in nodes to detect 6LoWPAN attacks, sending only flagged events to the edge for low overhead [17]. Effective security requires coordinated controls across devices, edge, and cloud layers.

TABLE I. SECURITY CONCERNS OF A THREE-LAYER IoT ARCHITECTURE

IoT Layer	Primary Role	Representative Threats	Typical Countermeasures
Perception	Sensing/actuation, data origination	Node capture, sensor spoofing, physical tamper [16]	Device authentication, tamper seals, secure boot
Network / Edge	Connectivity, preliminary analytics	DoS, MitM, routing manipulation [30]	Link-layer encryption, edge IDS, rate limiting
Application / Cloud	Storage, visualization, decision logic	Code-injection, malware, data breaches [30]	Application firewalls, access control, cloud IDS

B. Intrusion Detection Approaches

IoT intrusion detection relies on four paradigms. Signature-based systems match known threats, yielding low false positives but missing zero-days [16]. Anomaly based detectors like N-BaIoT spot deviations and novel attacks but with higher false alarms [19]. Specification based methods define rules for allowed actions, flagging violations, as in cyber-physical systems [18]. Hybrid approaches combine methods, balancing coverage and efficiency [25].

Supervised detectors need labeled data, while unsupervised or one-class models like incremental PCA [31] train on benign samples when labels are scarce. Choice depends on data, false alarm tolerance, and hardware limits.

TABLE II. TRADE-OFFS AMONG IOT INTRUSION-DETECTION PARADIGMS

Approach	Known-Attack Detection	Unknown-Attack Detection	False-Alarm Tendency	IoT Deployment Notes
Signature	Excellent [16]	None	Low	Requires ruleset updates, usually on gateway/cloud
Anomaly	Moderate	High [19]	Medium-High	Can be trimmed for on-device use
Specification	Moderate	High [18]	Low-Medium	Needs exhaustive rules; suited to well-specified CPS
Hybrid	High	High [25]	Medium	Balanced accuracy vs. cost; two-stage filters are common

C. Lightweight AI/ML

Most IoT nodes offer only kilobytes of memory and milliwatt-level power, so IDS models must apply Tiny ML principles to minimize resource use [22]. Feature selection (e.g., Chimp Optimization) drops redundant traffic attributes, reducing load while sometimes improving accuracy [27]. Pruning eliminates low impact weights, shrinking networks by 65% with little loss [22][32]. Quantization maps 32-bit weights to 8-bit or binary, yielding up to 128× memory savings and faster inference [20], [28]. Knowledge distillation compresses large models into compact students e.g.; a distilled transformer retained 95% accuracy at one-tenth size [23]. Online learning (e.g., CAEAID) enables adaptive updates without full retraining [24]. Lightweight ensembles combine shallow learners for improved recall under tight constraints [21]. Federated learning (FL), especially when compressed, enables privacy-preserving training across devices without central data transfer [26], [33].

TABLE III. COMMON OPTIMIZATION TECHNIQUES FOR LIGHTWEIGHT IOT INTRUSION-DETECTION MODELS

Optimization Technique	Primary Goal	Typical Savings	Representative IoT-IDS Example	Ref.
Feature selection/dimensionality reduction	Remove redundancy	↓ CPU & memory	Chimp-Optimized feature subset	[27]
Model pruning	Delete low-impact weights	~65 % size drop	Pruned CNN IDS	[22]
8-bit / binary quantization	Lower precision math	4×–32× smaller	BinaryConnect-style layers, edge-quantized IDS	[20], [28]
Knowledge distillation	Teach tiny model	×10 size reduction	Transformer student network	[23]
Online/incremental learning	Adapt to drift	No full retrain	CAEAID adaptive IDS	[24]
Lightweight ensembles	Boost recall cheaply	Minor extra RAM	Stacked pruned trees + KNN	[21]
Federated learning + compression	Train without raw data	↓ bandwidth	Optimal FL IDS for heterogeneous nodes	[26], [33]

D. Differences from General Endpoint Security

IoT security diverges sharply from traditional desktop or server protection across several fronts. Microcontrollers offer only kilobytes of RAM and low-MHz CPUs, requiring compressed or proxy defenses instead of full agents [22]. Battery constraints make energy-aware, quantized inference essential [28], while physical control roles demand sub-second detection for safety critical response [18]. Heterogeneous hardware and proprietary protocols necessitate protocol aware IDSs, unlike the standardized stacks in IT systems [30]. Scale and dispersion drive the use of federated or hierarchical architectures spanning device, edge, and cloud [26]. Field deployment adds physical tampering risks, as seen in SVELTE’s route anomaly detection through on-node monitoring [17].

TABLE IV. KEY CONTRACTS THAT NECESSITATE IOT-SPECIFIC INTRUSION-DETECTION STRATEGIES

Dimension	Desktop / Server Endpoint	IoT Device	IDS Design Implications	Refs.
Compute / memory	Multi-core, GBs of RAM	kB–MB RAM, low-MHz MCU	Aggressive compression and off-loading	[22]
Power source	Mains	Battery / harvested	Energy-aware inference, duty cycles	[28]
Real-time impact	Data-centric	Safety-critical	Sub-second latency mandatory	[18]
Protocol stack	Homogeneous	Highly heterogeneous	Protocol-aware / specification-based IDS	[30]
Deployment scale	Dozens–hundreds of hosts	Hundreds–thousands of nodes	Federated / hierarchical detection	[26]
Physical exposure	Controlled site	Unattended field	Tamper-evidence, trusted boot	[17]

IoT’s scale, diversity, and tight resources demand distinct intrusion detection. Layered architectures map threats and defenses, while signature, anomaly, specification, and hybrid methods offer complementary strengths. TinyML techniques show accurate IDSs can run within kilobyte memory and milliwatt power shaping the evaluation criteria ahead [34].

III. METHODOLOGY OF COMPARATIVE REVIEW

We conducted a structured review (2015–2025) of lightweight IDSs using IEEE Xplore, ACM, SpringerLink, ScienceDirect, PMC, and NSF databases. Searches included “IoT intrusion detection,” “lightweight IDS,” and “federated learning IDS.” We selected peer-reviewed papers focused on AI-based IDSs for resource-constrained IoT that reported quantitative results. Only experimental studies from top venues were included. Logged metrics included accuracy, F1, false positives, latency, model size, energy, and real time capability. Models were grouped into anomaly based, supervised lightweight classifiers, and federated/distributed IDSs, with TinyML and distillation outliers noted. Table 1 outlines resource performance tradeoffs.

A. Anomaly-Based IDS Approaches

Unsupervised or one-class learning enables anomaly based IDSs to detect deviations without signatures but requires compact models. Kitsune’s KitNET a cascade of micro autoencoders with online learning detects scans, MITM, and injections at 100 Mbps on Raspberry Pi [37]. N-BaIoT trains autoencoders for nine device types, nearly perfectly detecting Mirai/BASHLITE traffic but needs per-device models [38]. OC-AE Ensemble combines a Bi-GRU autoencoder with a lightweight ensemble for better interpretability without heavy classifiers [39]. Well-tuned autoencoders and one-class SVMs achieve 95–100% detection with low false positives. Efficiency comes from device-specific pipelines or feature partitioning; most models run on gateways, with few fittings on endpoints.

B. Lightweight Supervised and Tree-Based Approaches

Supervised IDSs often outperform anomaly detectors on well-structured data. Roy et al. used pruning, normalization, PCA, and class balancing with a lightweight B-Stacking ensemble, matching deep models on NSL-KDD and CICIDS2017 with <4% resource use [35]. Fenarir’s CART tree beat heavier models on KDD99 and UNSW-NB15, running on microcontrollers [40]. Alrashdi’s AD-IoT pruned a random forest to detect botnets efficiently in a smart-city lab [41]. Signature-based methods like Sheikh’s compressed matching for 6LoWPAN are efficient but not adaptive [42]. Alzahrani’s pruned CNN–LSTM achieved 92% accuracy, <0.4% false positives, under 6W on Raspberry Pi [36]. These models excel on known threats but miss novel ones, driving hybrid approaches for broader coverage.

C. Federated and Distributed IDS Approaches

Federated learning (FL) trains models locally and aggregates centrally for scalability and privacy. Lazzarini et al. deployed compact ANNs with FedAvg on ToN_IoT and CICIDS2017, nearing centralized accuracy with minimal communication; FedAvgM and FedAdam improved results further [43]. Zhang et al. added transfer learning to FL for industrial IoT, enhancing generalization across diverse nodes [44]. TinyML IDSs on microcontrollers show promise for ultra-low power use. Hybrid designs pairing gateway detectors with local monitors to reduce false positives while staying lightweight. These architectures balance performance and deployability in constrained settings.

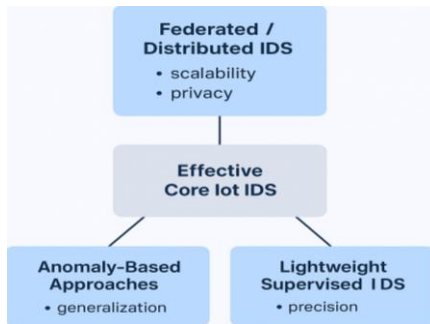


Figure 2 ML IDS Model Comparative Analysis

Overall, effectiveness relies on compact models, optimization, and deployment layers. Anomaly detectors offer generalization, supervised classifiers provide precision, and FL improves scalability and privacy. Table 5 summarizes models by technique, performance, and deployment, illustrating resource–security trade-offs in IoT IDS design.

TABLE V. SUMMARY OF COMPARATIVE STUDIES.

Model (Year)	Approach & Techniques	Detection Performance	Resource Footprint	Deployment Feasibility
Kitsune (Mirsky et al., 2018)	Unsupervised anomaly detection with tiny auto-encoders (KitNET)	High attack detection with low false-alarm rate after tuning	Runs in real-time on a Raspberry Pi; only a few MB of RAM	Edge-gateway NIDS monitoring up to 100 Mbps traffic
N-BaIoT (Meidan et al., 2018)	Deep auto-encoders model each IoT unit’s benign behavior	≈ 99 – 100 % TPR for Mirai & BASHLITE botnets; very low FPR	Small per-device models; lightweight runtime matrix ops	Gateway or cloud deployment; scalable to dozens of device profiles
OC-AE + Ensemble (Yao et al., 2023)	One-class Bi-GRU auto-encoder for zero-day detection plus label anomalies	> 95 % detection of unknown attacks; maps anomalies to nearest known class	Few hundred weights; inference in ms; total memory in low MBs	Suitable for IoT security gateways or industrial edge controllers
Decision Tree (DT) (Fenarir et al., 2019)	Supervised CART tree after filter-based feature selection	> 95 % accuracy on KDD99; outperformed SVM, KNN	< 100 KB model; negligible CPU/memory	Can run directly on microcontrollers or gateways
B-Stacking (Roy et al., 2022)	Lightweight stacked simple learners with PCA-reduced features	≥ 99 % AUC on CICIDS2017 & NSL-KDD; low FPR	~3 % RAM and ~2 % CPU on 8 GB host; small model file	Real-time on Raspberry Pi-class fog nodes
CNN-LSTM (Alzahrani et al., 2024)	Pruned hybrid CNN for spatial patterns + LSTM	92 % accuracy, FPR < 0.4 %	Runs on Raspberry Pi 4 within a 6 W power budget	Fog-computing gateway; energy-aware design
Federated ANN (Lazzarini et al., 2023)	FL with shallow ANN on each node; FedAvg, FedAdam variants	~97 % accuracy—nearly centralized baseline	Local models are small; updates tens of KB per round	Distributed IoT nodes + central coordinator; privacy-preserving
BT-TPF (Wang et al., 2024)	Knowledge-distilled PoolFormer transformer	98 – 99 % accuracy; matches teacher	≈ 10 % of original parameters; few MB model	High-end IoT gateways or edge servers; training done offline

Table 5 highlights the trade-off between detection and efficiency. Anomaly-based models like Kitsune [37] and N-BaIoT [38] spot novel threats on constrained devices but need tuning to reduce false positives. Supervised methods (e.g., Fenarir’s tree [40], Roy’s ensemble [35]) achieve high accuracy

but rely on labeled data and struggle with unknown attacks. Federated learning offers privacy and scalability with coordination costs [43][44], while knowledge distillation helps compress complex models for IoT use. The next section examines trends, challenges, and how lightweight IDSs integrate into IoT and enterprise security.

IV. RESULTS AND DISCUSSION

A. Different Patterns and Insights of Lightweight AI IDS Models for IoT

Effectiveness of Simpler Models: Contrary to assumptions, simpler algorithms like decision trees, compact neural nets, and one-class models perform comparably to complex ones in IoT due to the regularity of IoT traffic. For example, thermostats exhibit narrow behavioral patterns easily captured by simpler models. While deep learning excels on high-dimensional data, its complexity yields limited benefits in resource-constrained IoT. Studies show lightweight models, after feature selection and data balancing, often exceed 95% accuracy, balancing performance and efficiency [45][46].

Importance of Feature Engineering: Unlike enterprise IDS relying on end-to-end deep learning, IoT IDS focuses on feature engineering and dimensionality reduction. Given IoT's predictable traffic, PCA and domain-specific feature selection target relevant packet fields to reduce computation while maintaining detection. Manually selected features like packet rates and error counts flag threats, shifting away from automatic feature learning. Hybrid approaches combining expert-driven selection with lightweight models have become standard [47][48].

Resource Utilization Achievements: Reviewed models demonstrate low resource use with strong detection. Roy et al.'s system uses just 3% memory and 2% CPU; Alzahrani's CNN-LSTM consumes only 6W on a Raspberry Pi, proving real world feasibility. This addresses manufacturers' concerns about runtime overhead. Open-source models like Kitsune and N-BaIoT enable local resource testing. Raspberry Pi serves as a baseline for experiments, relevant even to lower-spec devices like Pi 3/4 [49][50].

Adaptability and Online Learning: IoT environments evolve with new devices and updates. Many lightweight IDSs use online learning or periodic retraining; Kitsune continuously updates its baseline. However, online adaptation risks poisoning attacks [51]. Federated learning helps by aggregating new data. A major challenge is adapting to benign changes (e.g., traffic spikes) without ignoring slow attacks, highlighting the need for concept drift detection an open problem in IoT IDS compared to non IoT contexts.

False Positive Rate: Managing false positives remains a major challenge, as excessive alerts cause alert fatigue. Anomaly based IDSs are especially prone if normal behavior isn't well

learned or benign anomalies occur. Though studies report 1% to 5% false positive rates, these come from controlled data, and real deployments struggle to maintain such levels. Mitigation includes multi-detector consensus, small alert thresholds, and retraining with feedback. Yao et al.'s ensemble method reduces alarm fatigue by mapping anomalies to known attack types [50][51][52].

Integration with General Endpoint Security: IoT devices, as specialized endpoints, are key to overall security. Malware detection techniques from PCs are adapted for IoT, using sequence models like LSTMs and transformers to detect complex multi-stage attacks akin to enterprise APTs. IoT IDS can also integrate with enterprise SIEMs for unified alerts. Federated learning allows collaborative improvement between IoT and traditional systems [50][52][53].

B. Challenges

Data Scarcity and Evaluation Realism: Building realistic IoT attack datasets remains difficult. Many studies still rely on legacy datasets like NSL-KDD or CICIDS variants, which poorly reflect IoT-specific traffic. While newer datasets like BoT-IoT and IoT-23 exist, they come with limitations [53]. Alternatives like shared data, IoT honeypots, and testbeds simulating hundreds to thousands of devices are needed to better assess scalability and real-world IDS performance.

Security of the IDS itself: IoT IDSs are also vulnerable. Adversaries can poison anomaly detectors by flooding traffic to shift baselines or compromise federated systems by corrupting a few nodes to taint the global model. Lightweight models, while more interpretable, are easier to reverse engineers. Research into adversarial resilience making models robust to poisoning is still emerging. Approaches like secure aggregation in FL or robust outlier filtering [52] offer potential protection.

Deployment and Maintenance: Deploying IDSs on IoT devices presents challenges in updating models, managing alerts, and avoiding disruption to device operations. One solution is to use edge security nodes to handle IDS tasks independently of the devices. For instance, a smart home hub can route all traffic and run the IDS, preserving device performance while maintaining monitoring [54].

C. The Future of IoT IDS Incorporation

TinyML IDS on Microcontrollers: Optimized models can now run directly on microcontrollers, detecting anomalies like sensor irregularities or suspicious command sequences locally bringing security to devices that previously couldn't connect to a central hub [55].

Collaborative Systems and Swarm Intelligence: In dense IoT settings, devices can share simple alerts via device-to-device links to collectively detect abnormal behavior like multiple

sensors acting strangely verified through local aggregation. Zero Trust Architecture complements IoT IDS by enforcing stricter authentication or reducing network privileges for suspicious devices. Zero Trust engines can also trigger deeper anomaly detection when unusual activity is detected [56].

Energy-Aware IDS: For battery-powered devices like wireless sensor networks, IDSs may use scheduling to monitor only at specific times or energy harvesting triggers to run analysis when resources allow. This can be achieved by measuring IDS power consumption and proposing some adaptive techniques [57].

Cloud-Offloading and Edge AI: Some solutions split IDS logic lightweight pre-processing at the edge and heavier analytics in the cloud. With 5G, low latency links enable real time detection using cloud AI, while minimizing data transfer by sending only extracted features or alerts [58].

In conclusion, constrained AI can deliver high accuracy with minimal footprints. A layered approach combining anomaly detection, lightweight signature models, and cloud-assisted or federated learning offers strong, energy-efficient protection for IoT systems.

V. CONCLUSION

This review shows that lightweight AI-based IDS for IoT has evolved from isolated trials into a versatile toolkit for real deployments. By comparing anomaly-based methods, compact classifiers, and federated learning across detection accuracy, false positives, CPU, memory, and energy usage, we identify the optimal designs for different IoT contexts. Unsupervised detectors offer wide-ranging threat coverage with minimal hardware cost, while small signature-based models give fast responses to known malware. Federated learning enables ongoing model improvement while protecting data privacy, assuming communication cost is manageable. Practitioners are encouraged to combine anomaly sensors, lightweight classifiers, and periodically federated retraining to build energy-efficient, adaptive systems. Looking forward, key challenges include creating diverse, standardized benchmarks; dynamic orchestration of models based on device telemetry; and adding explainability to ultra-compact models for operator trust and quicker incident handling. This study not only summarizes the current best practices but outlines the next steps toward scalable, collaborative cybersecurity across the growing IoT landscape.

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